Navigating Knowledge: Connecting Concepts, Communities, and Careers

Christine Herbst, Ly Minh Ngoc (Audrey) Tran, and Temuulen Enkhtamir

Introduction

Wikipedia contains extensive information about computing, but its article-by-article format can make an unfamiliar field difficult to navigate. A beginner exploring artificial intelligence, for example, encounters many related concepts without necessarily understanding how they connect, which deserve attention, or how they relate to practical interests. Our project addresses this problem by organizing articles and their relationships into an interactive knowledge map, supplemented by Reddit discussions and job board postings.

Our intended audience is learners seeking an initial understanding of computing and emerging technologies. We ask how a topic expands through hyperlinks, which concepts occupy important network positions, and where apparently separate branches reconnect. We also examine whether structural importance corresponds to popularity, how attention changes over time, and how Wikipedia readership compares with online discussion. A career-oriented extension connects this exploration with occupations.

Visualization is appropriate because these questions involve relationships, distributions, and temporal patterns that are difficult to communicate through article lists. The views support complementary tasks: following a manageable learning path, inspecting connections, comparing importance with attention, and investigating trends. Together, they provide several ways to explore a field without requiring users to understand network analysis before beginning.

Method

We processed English Wikipedia dumps using Python, converting article text exported by WikiExtractor and hyperlink records into Parquet tables. Separate Python scripts parsed compressed SQL dumps for page links, link targets, pages, and redirects. We standardized titles, retained the main article namespace, resolved redirects, and removed duplicate links and unresolved destinations. PyArrow supported batch conversion, while DuckDB performed large joins with configurable memory limits and temporary disk storage.

Starting from Artificial Intelligence, we selected 10,000 articles across ten subtopics. Using scikit-learn, we generated TF-IDF vectors from article titles and up to 6,000 characters of opening text, ranking candidates by their similarity to the parent article (75%), seed article (10%), and global in-degree (15%). Selection alternated across branches to distribute articles among subtopics.

We constructed a directed graph containing hyperlinks between selected articles and calculated in-degree, out-degree, and PageRank. Using NetworkX, we detected Louvain communities on an undirected graph version and estimated betweenness centrality to identify

potential bridges. Directed breadth-first search measured shortest hyperlink distances from the seed. Detected communities and shortest-path distances were distinguished from assigned subtopics and original selection tiers.

We collected monthly Wikimedia pageviews for September 2025 through August 2026 for each article. CSV and JSON exports supplied article attributes, edges, adjacency lists, communities, tree structures, and temporal summaries to the interface. Validation checked graph size, duplicate records, month coverage, and consistency between monthly and annual pageview totals.

We collected r/computerscience data through the SocialCrawl API using Python, combining paginated posts from "new," comment trees from highly commented posts, and searches for thirty computing-related terms. Requests budgets and incremental saving supported resumable collection. After excluding media-only, URL-only, deleted, and removed content, we retained 604 posts and 1,051 comments, with comments drawn from 105 threads. The dataset spans May 2025-October 2026 and was exported as JSONL and CSV.

We also used Rahman et al.'s "Job Dataset for IT Professionals," containing 8,797 postings with job titles, descriptions, skills, experience requirements, and salary ranges. Trimming whitespace reduced 254 distinct titles to 246, and requirements were parsed into skill lists. Salary bounds were standardized to thousands, and each job title's mean salary was calculated by averaging midpoints of its postings' salary ranges.

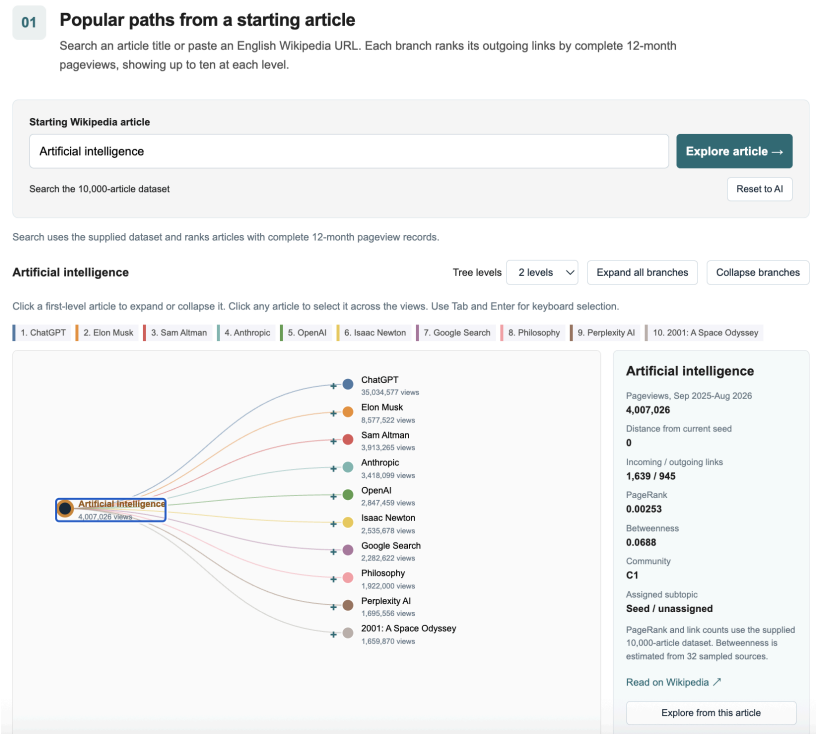
The interface uses HTML, CSS, JavaScript, and D3.js. Preprocessed CSV and JSON files support browser rendering, while consistent article identifiers enable coordinated selection, highlighting, and filtering across views.

Results

1 Seed Article Search and Hierarchal Tree

The tree establishes an entry point through an article title or Wikipedia URL. Its proposed structure displays the ten most popular connected topics, followed by another level of related topics. Users expand or collapse branches and limit depth, progressively revealing detail. It addresses how a field expands from a familiar starting concept and offers a manageable route through a much larger graph. Branching exposes candidate learning directions, although popularity-based selection can exclude less visible concepts.

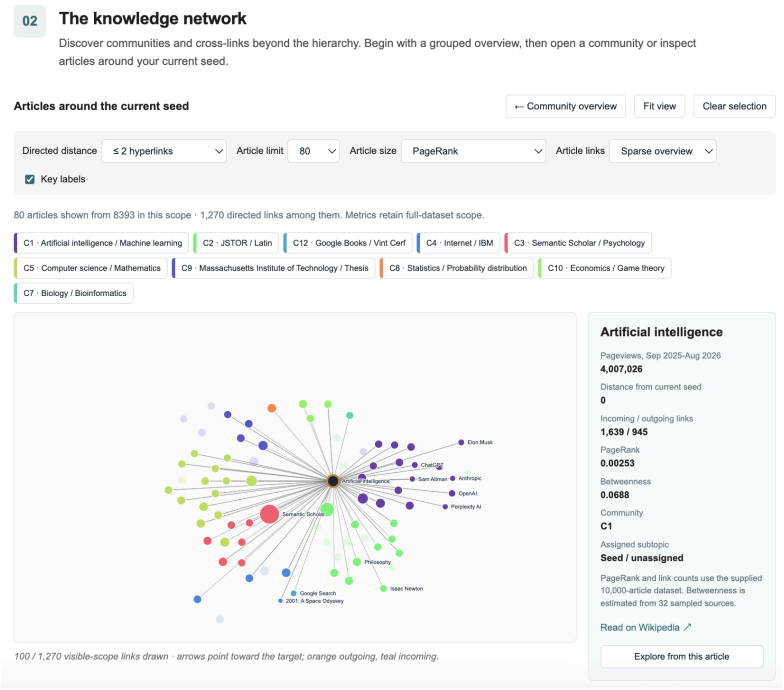
Figure 1.



2 Force Directed Knowledge Network

The network complements the tree by displaying relationships that a single-parent hierarchy cannot preserve. Articles appear as nodes and hyperlinks as edges; size and color can communicate structural importance and topic grouping. Zooming, panning, selecting nodes, and highlighting neighbors support local inspection. This view addresses hubs, bridges, clusters, and reconnections. It lets users investigate whether topics separated in the tree share articles or links.

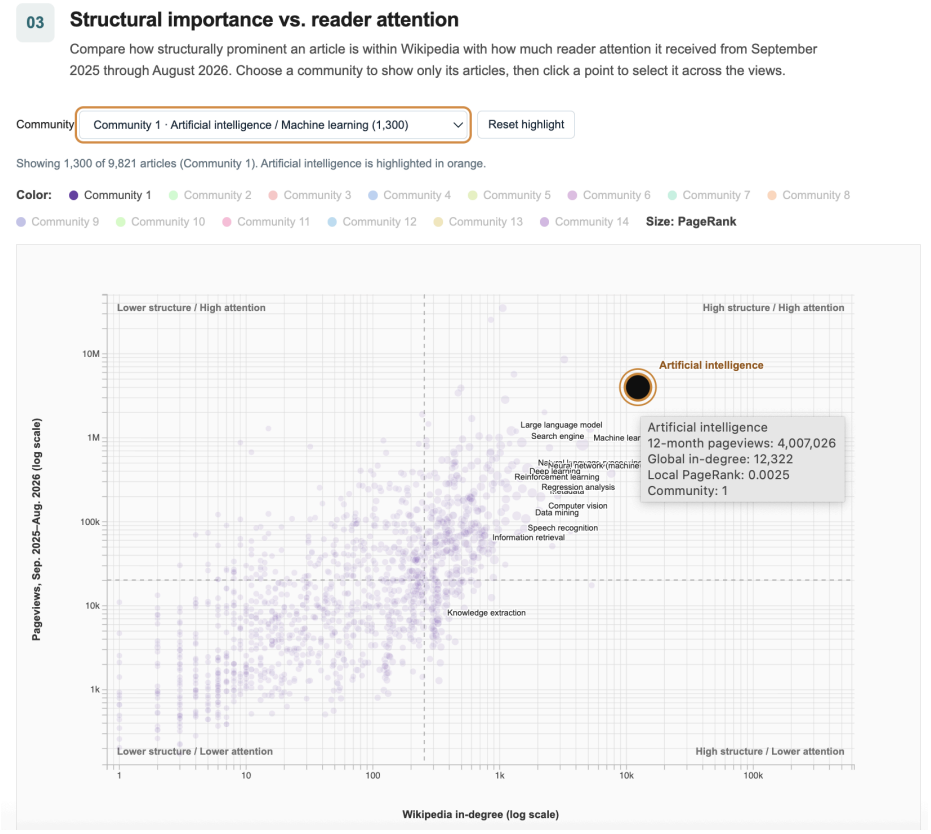
Figure 2.



3 Structural Important and Popularity Scatterplot

The scatterplot compares a network measure with Wikipedia pageviews, placing each article according to its structural position and reader attention. Quadrants help organize comparisons between relatively high and low values. Hover details and topic filters support identifying individual articles and investigating outliers. The question is whether highly connected or central concepts also attract substantial attention. Articles combining high importance with low attention are candidates for further exploration, but that classification depends on the selected measure and quadrant thresholds.

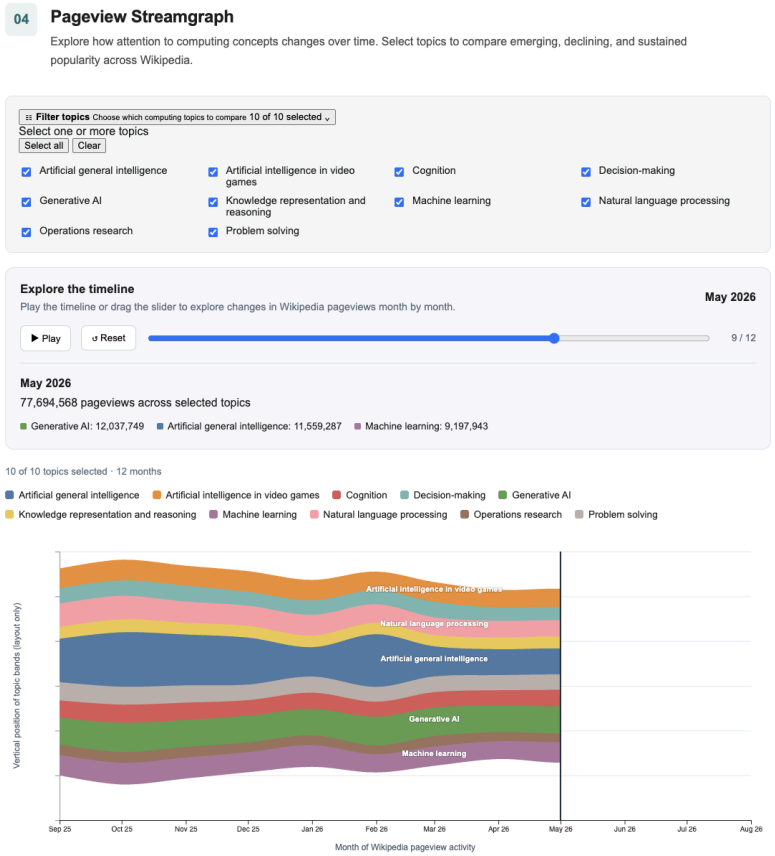
Figure 3.



4 Pageview Streamgraph

The pageview streamgraph represents attention over time through colored layers whose thickness indicates pageviews for selected concepts or groups. Topic selection and filtering help users compare sustained attention with temporary increases. It addresses whether interest is stable, emerging, or declining within the observation period. Broad layers indicate larger contributions, while changing thickness reveals temporal variation.

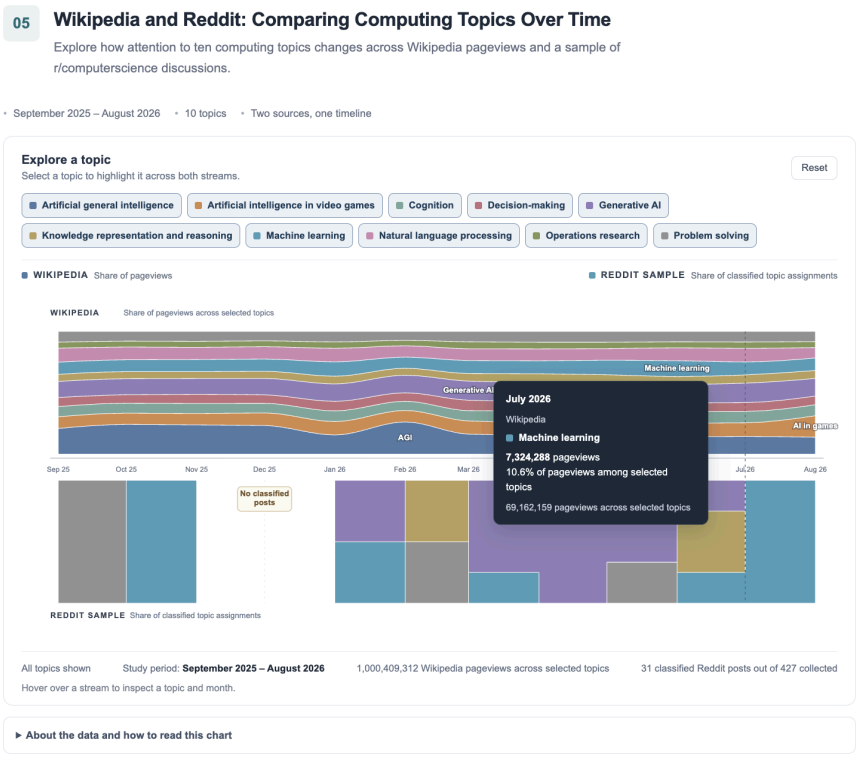
Figure 4.



5 Wikipedia and Reddit Comparative Streamgraph

The comparative view places Wikipedia attention above a central baseline and Reddit discussion below it. Corresponding concepts and aligned time intervals support inspecting similarities and differences between reading and discussion. Concept or subreddit filtering helps narrow the comparison. It supports investigating coincident peaks, possible timing differences, and concepts primarily in one source, subject to sampling and concept-matching limitations

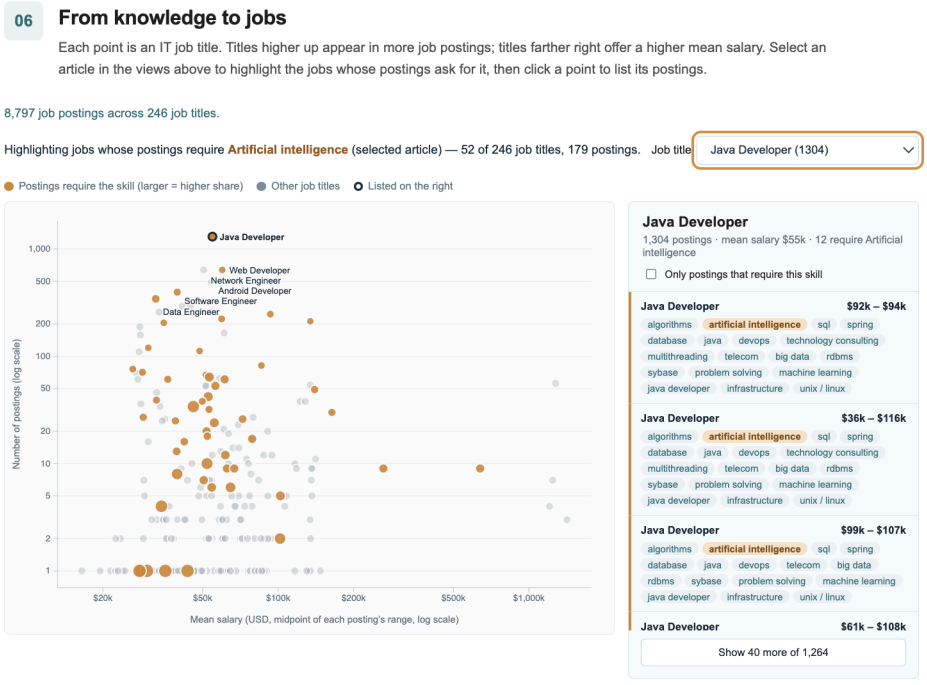
Figure 5.



6 Career Exploration Extension

The sixth view connects knowledge areas with occupations through a scatterplot mapping mean salary versus number of postings. Searching by job identifies associated topics, while searching by seed article identifies related occupations. It extends discovery toward practical learning decisions by helping users assemble areas to investigate for a career interest.

Figure 6.



Discussion

Our design combines complementary visualizations and interactions to help users explore computing concepts, compare their importance and popularity, and understand how they connect. The data includes articles, directed links, network attributes, temporal attention measures, discussion concepts, and occupation associations. The tasks include locating concepts, exploring relationships, comparing attributes, detecting outliers, and examining change. Trees and node-link diagrams represent structure; a scatterplot supports quantitative comparison; streamgraphs communicate changing composition. Search, filtering, selection, and details on demand reduce the amount of information users must process simultaneously.

A strength is the combination of complementary views. The tree supports a focused starting path, while the graph reveals relationships omitted by hierarchical presentation. The scatterplot separates different meanings of importance, and temporal views add context beyond a single popularity ranking. Consistent concept labels and coordinated selections can help users carry questions between these representations.

The same choices introduce weaknesses. Showing only popular neighbors makes the tree manageable, but biases exploration toward already visible articles. A force-directed graph can become a dense tangle, and layout movement can disrupt the user's mental map. Cluster filtering, selective labels, restrained edge opacity, and stable positions would improve readability. Selection should retain enough surrounding context to avoid making a local neighborhood appear disconnected from the broader field.

Data limitations also constrain interpretation. Wikipedia hyperlinks are editorial associations, pageviews measure attention rather than understanding, and selected Reddit communities represent a limited discussion sample. Concept matching may merge distinct meanings, or miss synonyms. Improvements should prioritize concise annotations, metric definitions, visible quadrant thresholds, and a clear explanation of the Reddit sample size. Preserve network context during highlighting, stabilize layouts, reduce edge clutter, and check color accessibility. Future work could manually audit Reddit classifications and report accuracy against a reliable reference set.

Conclusion

The project reframes computing knowledge as something to explore through several complementary lenses. The tree offers a starting path, the network reveals relationships beyond a hierarchy, the scatterplot distinguishes structural prominence from reader attention, and the two streamgraphs show how attention and classified discussion are distributed over time. The key interpretive lesson is that these measures answer different questions: network structure is not popularity, pageviews are not understanding, and a small Reddit sample is not a measure of public opinion. Used together, the views help learners generate better questions, discover connections worth following, and decide what to investigate next. The career extension adds a practical route

from knowledge areas to occupations, while requiring users to treat job-posting summaries as exploratory evidence rather than guarantees.

Roles and Responsibilities

Christine Herbst collected and cleaned all Wikipedia data used in the project, completed Visualizations 1 and 2, covering the search and hierarchical tree and the force-directed knowledge network, and was the primary coder for the interim check-in, and primary writer for the final report. Temuulen Enkhatmir collected and cleaned Reddit data and job board data used in the project, as well as completed Visualizations 3 and 6, covering the structural-importance versus popularity scatterplot and the career connection map. Audrey was the primary writer for the proposal and interim check-in, did finishing touches on the final report, and completed Visualizations 4 and 5, covering the pageview streamgraph and the Wikipedia-Reddit comparative streamgraph. All three members completed the poster.